

COMP9517: Computer Vision

2026 T1 Lab 4 Specification

Maximum Marks Achievable: 2.5

This lab is **worth 2.5% of the total course marks**.

Deadline for submission is Week 7, Monday 30 March 2026, 18:00 (Sydney time).

Objective: This lab revisits important concepts covered in the Week 5 lectures and aims to make you familiar with implementing specific algorithms.

Software: You are required to use OpenCV 3+ with Python 3+ and submit your code as a Jupyter notebook (see coding and submission requirements below). In the tutor consultation session this week and next, you can ask any questions you may have about this lab.

Materials: The images to be used in this lab are available via WebCMS3.

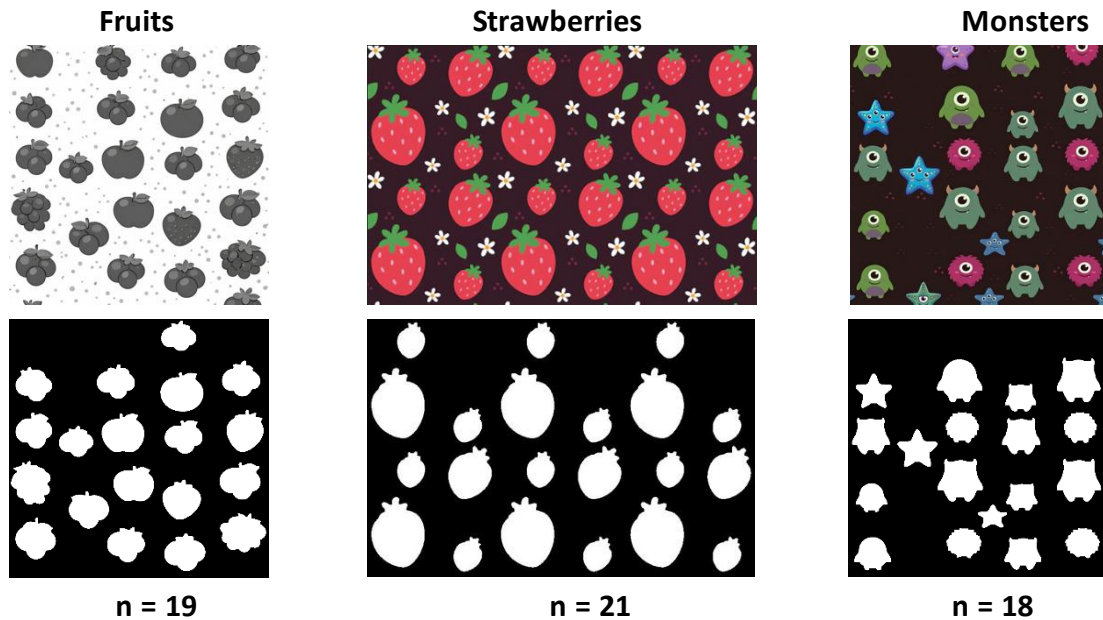
Submission: All code and requested results are assessable after the lab. Submit your source code as a Jupyter notebook (.ipynb file) that includes all outputs by the above deadline.

Task (2.5 marks)

In many computer vision applications, image segmentation is a crucial first step towards quantitative image analysis. As discussed in the lectures, there are many possible image segmentation methods, ranging from very simple to very complicated.

Despite its simplicity and limitations, intensity thresholding is still a very popular image segmentation method. This is because imperfections in the segmentation results can often be filtered out by postprocessing using binary morphological operators.

The goal of this lab is to write an algorithm that performs image segmentation and subsequent processing using just intensity thresholding and binary morphological operators. More specifically, the algorithm should be able to take any of the given lab images, produce a segmentation result containing only the bigger objects that are completely contained in the image, and print the number of objects (see the example results below).



Hints: Convert the input image to grayscale and apply Otsu intensity thresholding. Next, use opening and closing operators to filter out noise and fill holes in the segmentation, identify connected components and use reconstruction operations to get rid of segmented objects touching the image boundary, remove objects that are smaller than a certain size, and then count and print the final number of objects. Refer to Week 5 lecture slides Part 2 for descriptions of these morphological processing methods.

The above example results were produced by a single algorithm. That is, the same functions were used for processing each image. The only differences in execution for each image were the parameter values in some steps of the algorithm.

Coding Requirements

Obtain the segmentation results using a single algorithm (not a different set of functions for each image). The intensity threshold should be automatically determined. If your algorithm involves user parameters, their values must be specified clearly as input arguments, so that users can easily change and experiment with these parameters. Show the segmentation outputs in your Jupyter notebook as well as the count of objects (see examples above).

Make sure that in your Jupyter notebook, all cells in your notebook should have been executed so that the tutor/marker does not have to execute the notebook again to see the results.

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